

AGEL J020613–011417 — Drafting Fact Sheet for the Methods & Results Sections

Facts, decisions, and provenance keyed to the paper outline (2026-05-29)

Compiled for R. Córdova Rosado from project notes, memory, METHODS_AND_SYSTEMATICS.md, and TESTS_AND_DIAGNOSTICS.md

2026-05-29

How to use this document

This is a drafting reference, not prose for the paper. Every bullet is a fact or decision we have already made, with its source file in [brackets] so you can verify it. Where a number is superseded, the current value is given and the stale one flagged. Three gaps are flagged up front — they are not in this repo and you must source them elsewhere before drafting those paragraphs:

- G1 — Lens modeling (Ferrami et al.). No BPL / point-mass / $M(<r_{\text{break}})$ / joint-posterior / PSF / shear / subhalo content exists in this kinematics+photometry repo. Source from the lens-modeling repo (7 Documents/AGEL/202509_DESJ0206_modeling/ or 20251112_DESJ0206_Pyauto_PRONTO/) or the Ferrami draft directly.
- G2 — X-ray / “truly quiescent” statement. No X-ray, Chandra, or explicit quiescence data anywhere in the read files. The galaxy is only ever described as a “passive elliptical.” You need an external source if you want the X-ray quiescence claim in §Results.
- G3 — “We do not account for peculiar velocities.” This sentence is not currently written in any methods doc. It is a true statement about our analysis (we use the line-fit systemic z directly) — just add it; nothing to reconcile.

Headline numbers (current, 2026-06-08, post-M12; emitted by `scripts/paper_values.py` → `results/PAPER_VALUES.json`):

- $\sigma_e(<R_e) = 269.62 \pm 13.27$ km/s (asym $-13.45 / +13.10$); often rounded 270 ± 13 km/s. 2026-06-08: was ± 11.77 ; the D7 R_e -source systematic (± 6.13) was folded into the budget (M12, Task 5, nb15).
- $\log(M_{\text{?}}/M_{\odot}) = 11.16 \pm 0.08$ (stat) $+0.31$ (sys) at 10% flux errors [11.04 ± 0.14 (stat) $+0.32$ (sys) at 20%] — NEW principled arc masking (F200LP-located + IR-extended, raw aperture). The one-sided +sys is the Sérsic fill-in of deflector light hidden under the arc, reaching $\log M_{\text{?}} \approx 11.46$. Supersedes $11.33 +0.07/-0.09$ (older expert-aperture value, smaller masks; now a mid-range cross-check). See §2.1.1b / §3.2 and NOTES_photometry_mask_systematics_2026-05-29.md.
- $R_e = 2.305'' = 16.23$ kpc.
- $z_{\text{deflector}} = 0.67564$; $z_{\text{source}} = 1.302$.

Cosmology: FlatLambdaCDM, $H_0 = 70$ km/s/Mpc, $\Omega_m = 0.3$; $1'' \approx 7.04$ kpc (7043.5 pc/arcsec), $D_A = 1452$ Mpc at $z = 0.67564$. [reference_hst_jwst_data_properties.md]

Staleness warning: the *body* of METHODS_AND_SYSTEMATICS.md (dated 2026-05-18) still prints pre-M11 numbers (254.85 / 268.98, total ± 18) in its narrative prose. Its own §0 table and CLAUDE.md carry the current headline 269.62 ± 13.27 (post-M12, 2026-06-08). Always quote the current value (canonical source: `results/PAPER_VALUES.json`).

§2 Methods

§2.1 Photometric Analysis

§2.1.1 Masking

- Use the F200LP `_mask.fits` as the arc mask — NOT the F140W one. [feedback_masking_strategy.md; reference_hst_jwst_data_properties.md; METHODS §II.4]
 - F200LP mask (2512 HST pixels) is hand-tuned to cover the arc only and leaves the deflector core unmasked — it is the original arc-tuned segmentation mask.
 - F140W mask (1121 HST pixels) covers the arc PLUS the deflector core (the photometry tool’s aperture geometry pulled the core in); the two masks have zero spatial overlap.
 - Consequence: F140W is valued for its *image* (sharper Sérsic fit), never for its mask. Using the F140W mask as an arc mask would destroy the deflector-core signal. ### §2.1.1b Principled (reproducible) arc masking — NEW 2026-05-29

The hand-painted masks are now backed by an objective, reproducible recipe (so a referee can reproduce the arc selection), validated + applied to all four bands. [NOTES_arc_mask_verification_2026-05-29.md, NOTES_photometry_mask_systematics_2026-05-29.md; scripts/arc_mask_verification.py, mask_method_comparison.py, photometry_systematics.py]

- Locator = F200LP (highest contrast on the blue $z=1.302$ source). Two independent objective selectors reproduce the F200LP hand mask: (i) a color cut `m_F200LP–m_F140W` (arc bluer than the red deflector); (ii) a Sérsic-residual mask (subtract a 2D deflector model, flag positive excess $> 3\sigma_{\text{sky}}$). On F200LP the Sérsic-residual mask reproduces the expert photometry to 0.016 mag, R_e to 3%.
- Transfer to other bands by reprojecting the F200LP footprint (WCS + `map_coordinates`). Reproduces the expert JWST aperture mags to 0.01–0.02 mag. Do NOT use an independent per-band Sérsic-residual mask on F140W/JWST — a single Sérsic under-fits the bright extended IR galaxy (median residual +1.0) and over-masks by 0.2–0.4 mag.
- IR extends the source footprint: the deeper/sharper JWST bands see the arc’s fuller extent (F150W2 mask grows $\sim 6\times$ vs HST). We grow the F200LP arc into each band’s 2-component-Sérsic residual source pixels that are contiguous with the arc (region growing; cannot grab the core pedestal or field companions).
- Deflector model = 2-component (bulge+disk) Sérsic — single Sérsic under-fits JWST; 2-comp cuts galaxy-body RMS 3.3–3.5 $\times$. PSF effect on the fill quantified (`scripts/psf_fill_model.py`): a PSF-convolved 2-comp Sérsic (Gaussian at instrument FWHM; env lacks `webbpsf`) shifts the filled mag by ≤ 0.004 mag $\rightarrow \Delta M \approx 0.01$ dex, negligible. The fill is PSF-robust (arc is outside the PSF core).
- raw vs fill-in: raw photutils (masked) discards the deflector light under the arc; the Sérsic fill-in (replace masked pixels with the model) recovers it. With the large IR-extended masks the fill-in correction is +0.18 to +0.96 mag per band. filled photometry is mask-definition-independent (per-band vs global agree to 0.01 dex); raw is mask-size- dependent (up to 0.78 mag) — see §3.2.
- per-band vs global mask (union of all per-band masks): negligible for filled (0.01 dex $M_{\text{?}}$), large for raw — confirming filled is the robust quantity, raw the conservative lower bound.
- Files (in sibling dir `../velocity_dispersion_from_IFU/`): `AGEL020613-011417A_F200LP_WFC3_cutout_L3_mask.fits` (500 $\times$ 500, 25'' $\times$ 25''); the mask was revised 2026-02-28 (`_mask_20260228.fits`); original kept as `_mask_OLD.fits`.
- JWST bands: same arc geometry; the HST F200LP mask is re-projected from the HST WCS onto each JWST frame. [METHODS §II.4]
- For the IFU/ σ_e application (context): F200LP mask reprojected to the 0.30'' IFU grid via `scipy.ndimage.map_coordinates(order=0) nearest-neighbor`; $\sim 38/184$ spaxels inside R_e flagged ($\sim 21\%$ by count, $\sim 27\%$ by I-weight). [feedback_masking_strategy.md]

§2.1.2 Estimating R_e

- Headline $R_e = 2.305'' = 16.23$ kpc at $z = 0.67564$. [CLAUDE.md; METHODS §II.3/A3]
- Method: mean of the F140W and F200LP masked curves-of-growth (scripts/final_sigma_e.py: curve_of_growth, the PRODUCTION script):
 - F140W masked CoG = $2.168''$; F200LP masked CoG = $2.441''$; mean = $2.305''$.
 - Masked CoG zeros mask-flagged HST pixels before the radial flux integral, so the lensed arc / companions do not bias the half-light point.
 - Centers: HST-mean photutils.centroid_2dg centers (same as the σ_e pipeline), both bands.
- Do NOT confuse with scripts/measure_Re.py → results/Re_measurements.npz. That script uses the image-geometric center and produces 10 variants (3 sources $\times$ 4 masking strategies: unmasked, zeroed, proper, PSF-convolved); its “proper-mask” mean ($2.633''$) is not the headline — early-reference only. (It also hardcodes pixscale=0.08 for both HST bands, another reason production is preferred.)
- Superseded prior value: IFU white-light $R_e = 2.61''$ (nb05/06 era) — superseded because the HST-based $2.305''$ has the F140W+F200LP arc masks built in. A Ca H+K+G depth-map gives $2.866''$.
- R_e as a σ_e systematic (test D7): four R_e definitions shift narrow-window σ_e by up to 16.9 km/s (σ_e rises monotonically with R_e). FLAG: this component is currently held OUTSIDE the formal wide-window budget — flagged for a wide-window re-measurement. [CLAUDE.md]

§2.1.3 Measuring Fluxes

- Tools: photutils aperture photometry (interactive masking GUIs scripts/photometry_masking_{HST,JWST}.py). synphot is NOT currently used — flux→AB is done directly from header zeropoints. (*If the outline wants synphot, that is aspirational; flag with co-authors.*) [METHODS §II.4]
- AB zeropoints — read per-image from FITS headers, never hardcoded. [CLAUDE.md; reference_hst_jwst_data_properties.m
 - HST (drizzled e-/s): $ZP_{AB} = -2.5 \cdot \log_{10}(\text{PHOTFLAM}) - 21.10 - 5 \cdot \log_{10}(\text{PHOTPLAM}) + 18.6921$.
 - JWST (i2d, MJy/sr): $ZP_{AB} = -6.10 - 2.5 \cdot \log_{10}(\text{PIXAR}_{SR})$.
- Four bands / instruments / programs:

Telescope	Instrument	Filter	Program (PI)	Drizzled scale	EXPTIME	ZP_AB	AB (aperture)
HST	WFC3/UVIS	F200LP	16773 (Glazebrook)	$0.0500''/\text{pix}$	600.0 s	27.344	22.613
HST	WFC3/IR	F140W	16773 (Glazebrook)	$0.0800''/\text{pix}$	597.7 s	26.446	19.134
JWST	NIRCam SW	F150W2	05594 (Mahler)	$0.03075''/\text{pix}$	1836.0 s	28.033	18.942
JWST	NIRCam LW	F322W2	05594 (Mahler)	$0.06301''/\text{pix}$	1836.0 s	26.475	18.604

[reference_hst_jwst_data_properties.md] - JWST pointing comes from the cluster program ACT-CL J0206.2–0114 (ACT DR5, Hilton et al. 2021); the deflector sits in that field. No PAM correction (drizzled). Drizzle pixel correlation: empirical noise $\sim 1.3\text{--}2\times$ the idealized CCD-equation value — motivates the 10% flux floor (§2.1.4). - 2D Sérsic verification given masking (the morphology cross-check; notebook 08): single-component 2D Sérsic per band (scripts/measure_Re.py framework + the 2026-05-11 bound-fix $n \in [1.0, 6.0]$, $\text{ellip} \in [0.0, 0.6]$, 3-init grid — prevented an $n=0.30$ flat-disk escape on F200LP), extrapolated to total flux:

Filter	AB aperture	AB Sérsic-total	Δmag	n	$r_{\text{eff}} (")$
F200LP	22.613	20.672	−1.94	1.42	2.49
F140W	19.134	18.282	−0.85	1.54	1.88
F150W2	18.942	18.154	−0.79	1.40	1.72
F322W2	18.604	17.633	−0.97	1.97	2.03

Aperture under-counts outer-profile flux by 0.8–1.9 mag, but the integrated-mass effect is only +0.07 dex (see §2.1.4 / Results). Caches: `results/sersic_total_photometry.npz`, `results/bagpipes_sersic_refit.npz`. [METHODS §II.6]

§2.1.4 SED Modeling with `bagpipes`

- Tool: `bagpipes` (Carnall et al. 2018, 2019); BPASS+CLOUDY templates (`Bagpipes` default). Notebook `02_streamlined_Bagpipes_SED.ipynb`. [METHODS §II.5; `reference_sed_fitting.md`]
- Model: exponential- τ SFH; Calzetti dust (A_V free, 0–2); free metallicity (0–2.5 $Z_\odot$); redshift prior (0.674, 0.676) bracketing the line-fit z . Sampler `Nautilus` ($n_{\text{live}}=400$), 500-sample posterior.

```
fit_instructions = {
    "redshift": (0.674, 0.676),
    "exponential": {"age": (0.1, 15.), "tau": (0.3, 10.),
                    "massformed": (1., 15.), "metallicity": (0., 2.5)},
    "dust": {"type": "Calzetti", "Av": (0., 2.)},
}
```

- Why a 10% fractional error floor on each band: a conservative, uniform systematic budget that covers both the formal photon noise and the drizzle pixel-correlation underestimate (empirical noise $\sim 1.3\text{--}2\times$ the idealized CCD-equation value). [METHODS §II.4/II.7; `reference_sed_fitting.md`]
- Systematic test to fill in masked pixels = the 2D-Sérsic-total refit: because the aperture excludes arc-masked pixels and truncates the outer profile, we fit a 2D Sérsic per band (§2.1.3), extrapolate to total flux, and re-run `Bagpipes` with identical priors. This recovers the flux missing from masked/truncated regions and bounds the mass bias. Result: $\log(M_{\text{[O]}}/M_\odot) = 11.40 \pm 0.11 / -0.15$ (Sérsic-total) vs $11.33 \pm 0.07 / -0.09$ (aperture) $\rightarrow$ aperture is biased low by ~ 0.07 dex (+16% in $M_{\text{[O]}}$), within the quoted uncertainty. [METHODS §II.6] NOTE (2026-05-29): this analytic-total refit is now superseded by the per-aperture 2-component Sérsic fill-in under the principled IR-extended masks (§2.1.1b / §3.2): the headline is the raw value 11.16 (10%) with a one-sided +0.31 dex fill-in systematic reaching 11.46. The 11.40 analytic-total sits inside that bracket.
- Posteriors (older expert-aperture, now a cross-check): $\log(M_{\text{[O]}}/M_\odot)=11.33 \pm 0.07 / -0.09$; mass-weighted age 5.10 Gyr [3.69–6.11]; $\tau = 0.68$ Gyr; $Z = 0.89 Z_\odot$; $A_V = 0.82$ mag; $z_{\text{phot}} = 0.675$ [0.674–0.676]. Per-band residuals: F200LP +4.7%, F140W –11.3%, F150W2 –8.0%, F322W2 +9.6% (mild single- τ -SFH NIR-slope tension; not catastrophic, within $\pm 0.07\text{--}0.15$ dex). [METHODS §II.5]
- Caches / figure: `results/bagpipes_sed_results.npz` (the file Figure 2 loads), `pipes/posterior/AGEL0206/0206_real.h5` (delete to refit), `results/bagpipes_sersic_refit.npz`.

§2.2 Spectroscopic Analysis

§2.2.1 Calibration — @Keerthi / Kaustubh

- Instrument: Keck II / KCWI, KCRM red arm, Red-Low (RL) grating, Medium slicer, 2×2 binning; dithered at PA $0^\circ/45^\circ/90^\circ$. [METHODS §I.1; `reference_kcwi_data_properties.md`]
- Cube (headline): `Nov17_2025_DESJ0206_RL_combined_mtwdo_icubes_wcs.fits` — the NEW `_mtwdo_` reduction (Master Twilight + Dome hybrid flats), K. R. Gupta’s latest re-reduction. The earlier `_no_mtwdo_` cube is the OLD reduction (second-reduction cross-check; co-aligned to $<0.002''$, differs only in flat-fielding; σ_e differs ~ 7 km/s — the “reduction-pass” systematic).
- Reduction chain: raw frames $\rightarrow$ `kcwidrp`; stacked with `ISMgas/kcwi/kcwiReduction` by K. R. Gupta (Kaustubh); Dec-29 RED frames pre-reduced by Yuguang Chen. (*Calibration prose is Keerthi/Kaustubh’s to write — these are the facts on hand.*)
- Cube geometry: shape (3317, 100, 100); 0.300'' spaxels; $\sim 30''\times 30''$ FoV; $\lambda = 5625\text{--}8941$ Å at $\Delta\lambda = 1.0$ Å/pix; BUNIT = FLAM16/arcsec².
- Three confirmed nights (totals ≈ 180 min RED + 198 min BLUE on source):

Night (UT)	Program (PI)	RED	BLUE
2024 Dec 29	U204	4×300 s	1×1320 s (excluded by reducer)
2025 Aug 30	K409 (PI TBD)	12×300 s	4×990 s
2025 Nov 17	U002 (Jones)	20×300 s	5×1320 s

FLAG: the cube header (DATE-OBS 2025-11-17, PROGPI Jones) describes only the first input frame — cite the multi-night combine, not the header. K409 PI and Aug-30 DIMM seeing still TBD. - Spectral resolution / LSF: $\text{FWHM}_{\text{inst}} = 2.355 \times \text{DISPSCAL}(0.2940) \times 1.0 \text{ \AA} = 0.692 \text{ \AA} \rightarrow R \approx 10000$ at the fit band; $\sigma_{v,\text{inst}} \approx 12.6 \text{ km/s}$ observed (7.5 km/s rest-frame) — the $\sim 270 \text{ km/s}$ dispersion is resolved by $\sim 20\times$. LSF robustness $\leq 0.83 \text{ km/s}$ over $\text{DISPSCAL} \times 0.5\text{--}2.0$. - Seeing: guide-camera FWHM = $1.27''$ (GUIDFWHM, Nov-17; conservative).

§2.2.2 Redshift Measurement

- Method — single/multi-line Gaussian fits on absorption (+emission) features in the integrated spectrum, on NIST air rest wavelengths (`scripts/redshift_verify.py`, notebook 04):
 - Ca K 3933.66, Ca H 3968.47 (fit as a doublet with shared width), G-band 4304.40, H δ , H β , plus the [O II] 3726/3729 doublet (shared z , σ , flux ratio). Per-line z inverse-variance combined. H γ excluded (not trusted); Mg I / Fe / Na D out of range or blended.
 - Deflector systemic $z = 0.67564$ (supersedes AGEL DR2 $z = 0.67511$; $\sim 95 \text{ km/s}$ offset, used only in old notebooks). [METHODS §I.2; `reference_redshift_verification.md`]
- Double verification with ppxf: the Gaussian line-fit z is independent of, and consistent with, the ppxf V_{sys} (ppxf returns z via $z = (1+z_0) \cdot \exp(V/c) - 1$; Cappellari 2023 eq. 5c).
- Source $z = 1.302$ (AGEL DR2), cross-checked via Fe II UV resonance multiplet + Mg II 2796/2803 in the KCRM blue-arm cube. (*Measured $z_{\text{source}} \pm \text{uncertainty} + \text{its cache are a [TBD] placeholder — confirm before quoting a source- z error bar.$*)
- Peculiar velocities: we do not correct for peculiar velocities — we use the line-fit systemic z directly (see gap G3; this sentence needs to be added to the draft).
- Wavelength convention: AIR rest wavelengths throughout (NIST-verified); KCWI native vacuum is converted to air. Do not mix SDSS vacuum values ($\sim 1 \text{ \AA} \approx 80 \text{ km/s}$ systematic at $4000 \text{ \AA}$). [`feedback_wavelength_convention.md`]

§2.2.3 Stellar Kinematics & Velocity Dispersion

Spatial regime / masking / R_e

- Headline σ_e is measured on a single $I(r)$ -weighted 1-D aperture spectrum at $R < R_e$ — the literature-standard σ_e (Cappellari et al. 2006 eq. 1; the KH13 / Greene+20 / SAURON / ATLAS3D / MaNGA convention). $R_e = 2.305'' = 16.23 \text{ kpc}$ (§2.1.2).
- Center: HST-mean `photutils.centroid_2dg` of F140W+F200LP cores, propagated through the KCWI WCS; adopted RA = 31.55613° , Dec = -1.23819° (ICRS). F140W $\leftrightarrow$ F200LP offset $0.36''$ (F200LP shifted by the UV-bright arc; F140W is the clean bulge centroid).
- Spatial arc mask: F200LP segmentation mask reprojected to the IFU grid (nearest-neighbor); 38/184 spaxels inside R_e flagged ($\sim 27\%$ by I-weight). F140W mask not used (covers the core).
- I-weight map: per-spaxel collapse of the IFU 6500–7500 $\text{\AA}$ white-light band; spectra summed (not averaged) so the wild bootstrap preserves per-spaxel noise.
- $R < R_e/8$ (JFK95 σ_c) was dropped — at $0.288''$ it sits inside seeing $\text{FWHM}/2 = 0.64''$ (~ 3 spaxels). No central σ is quoted; $\sigma_e(<R_e/2) \approx 225 \text{ km/s}$ is reported as a gradient diagnostic only.

ppxf setup

- ppxf v9.x (Cappellari & Emsellem 2004; Cappellari 2017, 2023), moments = 2 (first two velocity moments V, σ). [METHODS §I.5/I.6; `scripts/bootstrap_ppxf.py`]
- Polynomial degrees: additive Legendre degree swept over 15–29 (15 values) and pooled; `mdegree` = 0. σ -vs-degree is flat within the bootstrap envelope (no LOSVD absorption by the polynomial). (*Note: METH-*

ODS §I.5 prose mis-labels these “multiplicative” — the code passes $\text{degree}=\dots$, $\text{mdegree}=0$, i.e. additive. Describe them as additive.)

- Three SPS libraries, pooled at the posterior level: FSPS, EMILES, XSL (8–9 templates each over rest 3500–5000 Å).
- Per-SPS native wavelength frame: FSPS = vacuum, EMILES = air, XSL = air. Galaxy frame matched per-SPS (scalar-median vac↔air via Ciddor 1996, following Cappellari’s pattern). Why: the legacy “convert galaxy to air for all three” produced a spurious –90 km/s FSPS V_{sys} ; the fix collapsed the V_{sys} split-track $\sim 110 \rightarrow \sim 15\text{--}18$ km/s (σ shift ≤ 2 km/s). [reference_ppxf_vacair_handling.md]
- Per-SPS V_{sys} subtracted before pooling (σ is V-invariant; V_{sys} medians FSPS –19, EMILES –4, XSL –1 km/s). A single global V_{sys} would inflate σ_e by $\sim 10\text{--}15$ km/s in the discrete cross-check.
- Emission-line handling (sigma clipping / removing emission lines):
 - Custom no-Balmer goodpixels (`_determine_goodpixels_no_balmer`): masks only the forbidden lines ([O II], [O III], [O I], [N II], [S II]) and keeps H δ , H γ , H β IN the fit (they are absorption in this passive deflector; masking them discards ~ 120 stellar pixels). TODO flag: H δ may still need targeted masking — open item.
 - $z = 1.302$ source-emission masks (ARC_MASKS_REST, mapped by $(1+z_s)/(1+z_l)=1.374$): Mg II 2796/2803 (def-rest 3835–3855 Å), [O II] 3727/3729 (5115–5135), [Ne III] 3869 (5260–5340), He I 3819 (5237–5253, added 2026-05-27 as test M8). Plus an O₂ A-band telluric residual at def-rest 4525–4545 (not source — confirmed 2026-05-18).
 - M10 sky-line audit: bad-pixel catalog BAD_PIXELS_REST = 35 entries (26 CR residuals + 9 OH air-glow/sky bands), all verified non-source via the arc spectrum.
- Wild bootstrap errors: hybrid Rademacher sign-flip $\times$ local-residual scaling (75-pixel rolling-window residual rescale, clipped [0.2,5.0]); $N = 500$ production per (SPS $\times$ degree), $N = 50$ smoke (agree within 1 km/s). Errors = 16/84 percentiles (asymmetric). joblib parallel, BLAS pinned to 1 thread/worker.

σ_e definition & wavelength window

- σ_e definition (Gültekin et al. 2009; Kormendy & Ho 2013 eq. 3; Greene et al. 2020 ARA&A; 2D IFU form Cappellari et al. 2006 eq. 1):

$$\sigma_e^2 = \frac{\int_0^{R_e} (V^2 + \sigma^2) I(r) dr}{\int_0^{R_e} I(r) dr}.$$

The headline path is the cumulative I-weighted single-ppxf fit on the $R < R_e$ aperture spectrum (preserves LOSVD shape; matches what KH13/Greene+20 compute), not the discrete annular sum.

- Standard wavelength range (headline): rest 3800–5400 Å (`wR3800_5400_arcmask`) — Ca H+K through Mg b / Fe5270 / Fe5335; 2161 good pixels ($\sim 2.6\times$ the narrow window). The fit is constrained by the highest-S/N features — Ca H, Ca K, G-band (3800–4400) — and is consistent across windows.
 - Blue edge 3800 Å = XSL template floor (def-rest 3675 Å) + 5% velocity padding + retaining Ca H+K; red edge set by KCWI coverage. Provenance is data-driven, NOT a specific paper. Closest precedent: LEGA-C (Bezanson et al. 2018; van der Wel et al. 2021) at $z \sim 0.7$. DO NOT cite “Cappellari 2017 recommends extending blueward” — that was a hallucination, retracted.
 - Cross-check window: rest 4000–5400 Å (`wR4000_5400_arcmask`, H β + Mg b, no Ca H+K) — orthogonal Lick-tradition feature set.
 - Narrow legacy window: obs 6500–7500 Å = rest 3879–4476 Å (Ca H+K + G); $\sigma_e = 267.95 \pm 30.10$.

A few systematics tests to describe (full budget in Results §3.1): I-shape (10 I-weight-map shapes, ± 2.27), F200 mask weight ($w \in \{0, 0.5, 1\}$, ± 6.65), fit-window (3 windows, ± 3.82), frame (vac/air, ± 5.0), centering (5-center $\pm 0.4''$ sweep, ± 4.0), reduction-pass (NEW vs OLD cube, ± 3.45). Plus the M9 “DO NOT MASK” finding: the +5–7% bump at def-rest 5193–5204 Å is the Mg b LOSVD wing (signal, not contamination); masking it drops σ_e by 7 km/s.

§2.3 Lens Modeling

GAP G1 — not in this repo. The companion-paper (Ferrami et al.) lens-modeling content (multiple lens models, PSF construction, joint shear+subhalo constraints, potential-correction triple-checks, BPL vs point-mass differentiating models) is not present in this kinematics+ photometry repo. Source it from the lens-modeling repo (`7Documents/AGEL/202509_DESJ0206_modeling/`, `20251112_DESJ0206_Pyauto_PRONT0/`) or the Ferrami draft. The only on-hand pointer: `PROJECT_BRIEF.md` notes the lens-model enclosed mass combines with $(\sigma, M_{\odot}, z)$ to give a 4D constraint on the central BH + bulge scaling relations.

System identity for the summary sentence: AGEL J020613–011417 (DES J0206–01), AGEL DR2; source $z = 1.302$ (spiral), deflector $z = 0.67564$.

§3 Results

Figure 2 (left panel) — the integrated spectrum + SED

- Left: KCWI/KCRM Red-Low integrated, $I(r)$ -convolved spectrum of the deflector elliptical, with the ppxf best fit overlaid (and inline residuals — Rodrigo’s standard nested-gridspec style). Loads from the wide-window fit + bootstrap pool.
- Right: HST+JWST four-band SED + Bagpipes posterior model. Loads `results/bagpipes_sed_results.npz`.
- Figure file: `AGEL0206_sigma_e_SED_final_wide.pdf` (per outline)/`results/AGEL0206_spectra_SED_fit.pdf`.

§3.1 Velocity Dispersion Measurement

- Headline: $\sigma_e(<R_e) = 269.62 \pm 13.27$ km/s (asym $-13.45/+13.10$), via the Gültekin (2009) luminosity-weighted formalism evaluated with the Cappellari ppxf implementation on the $I(r)$ -weighted $R<R_e$ aperture spectrum, plus the wild-bootstrap error pool.
- Reproduce: `scripts/run_wide_sigma_e.py --cube new_clean_hei --n_bootstrap 500`. Caches: `results/run_wide_sigma_e/new_clean_hei/wR3800_5400_arcmask_{fsps,emiles,xsl}_T*_N500.npz`.

R_e measurement and checks (for the §3.1 paragraph): $R_e = 2.305'' = 16.23$ kpc, the F140W (2.168'') + F200LP (2.441'') masked-CoG mean; supersedes IFU-only 2.61''; masked CoG removes arc/companion bias.

Systematic error budget (post-M12, 2026-06-08; computed by `scripts/paper_values.py`):

Component	$\pm$ km/s	Why
stat (N=500)	4.6 ($-5.16/+4.13$)	wild-bootstrap pool over 3 SPS $\times$ 15 degrees; already marginalizes SPS (between-SPS ± 2.04) + degree spread
I-shape (10 shapes)	2.27	peak-to-peak/2 of 10 I-weight-map shapes (266.83–271.37)
F200 mask (3 weights)	6.65	(w00 269.69 – w100 256.39)/2; cleaned cube has more signal pixels $\rightarrow$ more arc-dilution leverage. (Arc-mask-definition cross-check ± 5.85 overlaps this, not added — M12/nb13)
frame (vac/air)	5.0	structural per-SPS native-frame choice
centering (HST WCS)	4.0	5-center $\pm 0.4''$ sweep; reduction-independent
fit-window (3 windows)	3.82	wR3800_5400 269.66 / wR4000_5400 268.58 / w6500_7500 276.22; dropped from carried ± 15 on the cleaned cube

Component	$\pm$ km/s	Why
reduction-pass	3.45	(269.62 – 262.72)/2, NEW vs OLD cleaned cube; only 2 reductions — refine if a 3rd lands
R_e-source (D7 wide)	6.13	NEW M12 (2026-06-08, nb15): peak-to-peak/2 over 4 R_e estimators (mean 2.305"/F140W 2.168"/F200LP 2.441"/CaHK+G 2.902" → 269.62/267.44/272.44/279.69). User chose full spread; CaHK+G's +10 is the rising $\sigma(R)$ gradient. Light-family-only = ± 2.50
TOTAL (sym)	13.27	quadrature
TOTAL (asym)	–13.45 / +13.10	preserves stat skew

- No separate SPS line is folded in: the SPS-library spread collapsed from ~ 26 km/s (narrow: FSPS 253.0 < EMILES 267.5 < XSL 279.7) to ~ 4 km/s (wide) — a key argument for the wide window; it now sits inside the pooled stat width.
- Quadrature is deliberately conservative (components not strictly independent).
- FLAG: R_e-source systematic (± 16.9 km/s at narrow window) is NOT yet folded in — flagged for a wide-window re-measurement.

Systematics paragraph order (per outline): first R_e and I(r) (I-shape ± 2.27 , R_e-source flagged), then the spectrum (mask ± 6.65 , frame ± 5.0 , window ± 3.82 , reduction ± 3.45 , centering ± 4.0).

Cross-checks (narrow window, architecturally independent, superseded for headline but valid):

- §6cum cumulative I-weighted aperture ppxf: 267.32 ± 24 ; $\sigma_e(<R_e/2) \approx 225.78$; $\sigma_e(<R_e/8) \approx 209.18$.
- §7 discrete Gültekin annular sum (equal-N inside $R_{\text{safe}}=3R_e/4$): $256.17 -13.0/+12.7$.
- §7b flat- σ extrapolation: $274.37 -16.2/+17.4$.
- nb07e arc-spectrum subtraction matches §6cum bit-identically (<1 km/s).
- Narrow Ca H+K + G window (nb09d): 267.95 ± 30.10 — headline within ~ 2 km/s.

Headline evolution (mask-audit arc): 268.98 (clean) $\rightarrow 271.87$ (+He I, M8) $\rightarrow 269.62$ (+M10 sky audit) $\rightarrow \pm 11.77$ (M11 re-derivation) $\rightarrow \pm 13.27$ (M12, +R_e-source ± 6.13 , 2026-06-08). He I pushes σ up ~ 2.9 , the 9 sky bands pull it down ~ 2.2 (near-cancel); central value 269.62 stable across M10 $\rightarrow$ M12 (only the error grew).

Compare with Greene (2016/2020) and KH13 choices: $\sigma_e(<R_e)$ here uses the Cappellari et al. 2006 eq. 1 luminosity-weighted single-aperture definition, exactly as adopted by Kormendy & Ho 2013 (M $\bullet$ - σ eq. 3) and Greene et al. 2020 ARA&A (fig. 5). We dropped $R < R_e/8$ (JFK95 σ_c) because seeing under-resolves it. For a pure elliptical, galaxy $R_e \approx$ bulge R_e , so KH13's "bulge R_e " equals the measured total R_e — no bulge/disk decomposition needed.

σ_e physical-meaning discussion: the deflector is a dispersion-supported elliptical bulge; $\sigma(r)$ should *decrease* gently with radius — any rising $\sigma(r)$ in raw output is a continuum-dilution artifact from the $z=1.302$ arc, not a physical gradient. Resolved kinematics (nb11): 17 central spaxels at $S/N \geq 5$, $\sigma \in [144, 251]$ km/s (median 201), declining with R , no coherent rotation above noise.

§3.2 Stellar Mass Estimate

- $\log(M_{\text{?}}/M_{\odot}) = 11.16 \pm 0.08$ (stat) $+0.31$ (sys) at 10% flux errors [$11.04 \pm 0.14 +0.32$ at 20%] — NEW headline from the principled F200LP-located + IR-extended arc masking with raw aperture photometry (scripts/photometry_systematics.py; results/photometry_systematics_Mstar.npz; fig results/figures/Mstar_budget.png). Empirical / conservative: raw masked photometry discards the deflector light under the (now IR-extended) arc, so the central is biased LOW and the systematic is one-sided UP.
- The systematic is intentionally uneven (could be higher): the Sérsic fill-in — which restores the deflector light under the masked arc and is mask-definition-independent (per-band vs global agree to 0.01 dex) —

gives $\log(M_{\text{?}}/M_{\odot}) = 11.46$ (10%) / 11.36 (20%), i.e. the $+0.31$ dex upper reach. The mask-definition term is negligible (± 0.003 – 0.034 dex).

- Flux-error choice matters at ~ 0.10 dex: 10% $\rightarrow$ 20% lowers the central ($11.16 \rightarrow 11.04$, looser likelihood lets the prior pull mass down) and widens stat ($\pm 0.08 \rightarrow \pm 0.14$). Both quoted.
- Explicit masking-approach systematic on $M_{\text{?}} = \pm 0.16$ dex (10%: ± 0.160 , 20%: ± 0.162) — peak-to-peak/2 across all masking approaches {expert-aperture, per-band/global $\times$ raw/filled} (`results/Mstar_masking_systematic.npz`, `fig Mstar_masking_systematic.png`). Decomposition: under-arc light (raw $\leftrightarrow$ filled) ± 0.15 (dominant; the adopted one-sided $+0.31$), mask-definition (per-band $\leftrightarrow$ global) ± 0.004 (negligible — filled is mask-def-independent), mask-extent (expert $\leftrightarrow$ IR-extended) 0.18 . This ± 0.16 is the symmetric statement of the same effect quoted one-sided ($+0.31$) in the headline.
- Superseded values (cross-checks, smaller masks): older expert-aperture $\log(M_{\text{?}}/M_{\odot}) = 11.33 + 0.07 / -0.09$ (sits mid-range); 2D-Sérsic-total refit $11.40 + 0.11 / -0.15$. Consistent with the new raw $\leftrightarrow$ filled bracket [$11.16, 11.46$].
- “Typical elliptical at $z \sim 0.7$ ”: $M_{\text{?}} \approx 1.4 \times 10^{11} M_{\odot}$ (raw headline) to $\approx 2.9 \times 10^{11}$ (fill-in upper) at $z = 0.6756$, passive/quiescent SFH (mass-weighted age ~ 5 Gyr, low sSFR). Place on the $z \sim 0.5$ – 1.0 mass-size relation (van der Wel et al. 2014; Mowla et al. 2019) and the strong-lens-deflector analog sample (Sonnenfeld et al. 2015, SL2S V — best z -match). [notebook 10 verified citations]
- X-ray / quiescence: GAP G2 — no X-ray data in repo. Source externally if you want the “truly quiescent” claim; otherwise frame from the passive SED only.

§3.3 Mass Estimates from Lens Modeling

GAP G1 — see §2.3. The range of lens-model results, the joint-posterior mass + uncertainty, the BPL vs point-mass differentiating models, $M(< r_{\text{break}})$, the pedagogical-figure intuition, the small table reproduced from Ferrami et al., the comparison to local galaxies, and the “shape of the mass profile matters” argument are all in the lens-modeling repo / Ferrami draft, not here. The only in-repo pointer is the 4D-constraint sentence in PROJECT_BRIEF.md. Pull these numbers from the companion paper before drafting §3.3.

Key citations (verified in notebook 10)

- ppxf: Cappellari & Emsellem (2004); Cappellari (2017, MNRAS 466, 798); Cappellari (2023).
- σ_e aperture definition: Cappellari et al. (2006, MNRAS 366, 1126) eq. 1; Gültekin et al. (2009) eq. 1; Kormendy & Ho (2013, ARA&A 51, 511) eq. 3 (scatter 0.29 dex); Greene, Strader & Ho (2020, ARA&A 58, 257) fig. 5; Saglia et al. (2016).
- $z=0$ IFS calibration: Cappellari et al. (2013, ATLAS3D XV & XX); Krajnović et al. (2013, XVII); Zhu et al. (2024, MaNGA DynPop III).
- Mass-size / analogs: van der Wel et al. (2014); Mowla et al. (2019); Ge et al. (2021); Sonnenfeld et al. (2015, SL2S V).
- Vacuum $\leftrightarrow$ air: Ciddor (1996). Wide-window precedent: Bezanson et al. (2018); van der Wel et al. (2021).
- Source- z catalog: AGEL DR2 (Carleton et al., in prep). Cluster: Hilton et al. (2021, ACT DR5).

Flagged open items (carry into drafting)

1. ~~R_e-source-systematic-not-folded-in~~ RESOLVED 2026-06-08 (M12, nb15): re-measured at the wide window (± 6.13) and folded into the budget $\rightarrow$ headline $\pm 11.77 \rightarrow \pm 13.27$.
2. Reduction-pass ± 3.45 rests on only 2 reductions — refine if a 3rd lands.
3. ~~H δ may still need targeted masking~~ RESOLVED 2026-06-08 (M12, nb16): decision = keep unmasked (H δ is well-fit, not a contaminant; masking it would discard LOSVD information). TODO in `bootstrap_ppxf.py` closed. 3b. NEW (A3c, nb14): raw CoG R_e is r_max-non-convergent; headline $2.305''$ is the top of a 2.1 – $2.5''$ family (R_e method systematic $\pm 0.08''$), not folded into $\sigma_e/M_{\star}$ — flagged for a decision.

4. Asymmetric-error tables: older 4-corner/M10 tables print $-17.92/+17.65$ (pre-M11), M11 was $-11.98/+11.57$ (± 11.77); current (M12) is $-13.45/+13.10$ (± 13.27). Quote the M12 value (canonical: `results/PAPER_VALUES.json`).
5. No X-ray / quiescent data in repo (G2). 6. No lens-model content in repo (G1).
6. “We do not account for peculiar velocities” needs to be written in (G3 — true, just add it).
7. Source-z (1.302) error bar + its cache are a [TBD] placeholder.
8. K409 (Aug-30) PI and per-night DIMM seeing still TBD for acknowledgements.